MindWorldBench: Evaluating Mental-State-to-Behavior Reasoning in Image-to-Video Generation

Ruiqi Li*
rqli25@stu.pku.edu.cn
Peking University

Xuanyi Liu*
xuanyi@stu.pku.edu.cn
Peking University

Sijia Li*
lisijia@xs.ustb.edu.cn
University of Science and Technology Beijing

Haofeng Wang
hfwang@stu.pku.edu.cn
Peking University

Yuxin Liu
u202442516@xs.ustb.edu.cn
University of Science and Technology Beijing

Feng Xie
u202442730@xs.ustb.edu.cn
University of Science and Technology Beijing

Songchao Tan
sctan@ustb.edu.cn
University of Science and Technology Beijing

Shiqi Wang
shiqwang@cityu.edu.hk
City University of Hong Kong

Hanwei Zhu
hanwei.zhu@ntu.edu.sg
Nanyang Technological University

Yizong Wang†
wang@pku.edu.cn
Peking University

Chuanmin Jia
cmjia@pku.edu.cn
Peking University

Siwei Ma
swma@pku.edu.cn
Peking University

Unpublished working draft. Not for distribution.

Abstract

Current image-to-video models achieve visual realism and physical plausibility, but reasoning about mental states remains unexplored. Actions are driven by belief, desire, and perception, requiring inference beyond explicit instructions. We introduce **MindWorldBench** to evaluate mental-state-conditioned video generation. We formalize this as mental-state-to-behavior reasoning, where models generate actions from a world state and latent variables without explicit action prompts. MindWorldBench utilizes **Zero-Action Prompting** and a counterfactual design with 744 prompts to isolate the causal effects of mental states. An automated pipeline evaluates video quality, commonsense plausibility, and mental-state consistency. Evaluations of eight video models show that despite visual fidelity and physical reasoning, models fail to align behaviors with latent mental states. We identify a failure mode, termed **Omniscient Bias**, where models default to the objective world state rather than human's subjective belief. These results demonstrate a disconnect between visual generation and cognitive reasoning, suggesting a need for explicit mental-state modeling in video generation systems. Project website: https://richard2049-lee.github.io/MindWorldBench/

*These authors contributed equally to this research.
†Corresponding author.

Unpublished working draft. Not for distribution.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference acronym 'XX, Woodstock, NY

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/2018/06
https://doi.org/XXXXXXX.XXXXXXX

CCS Concepts

• **Computing methodologies** → **Computer vision**; • **General and reference** → **Evaluation**.

Keywords

video generation, image-to-video, mental state reasoning

ACM Reference Format:
Ruiqi Li, Xuanyi Liu, Sijia Li, Haofeng Wang, Yuxin Liu, Feng Xie, Songchao Tan, Shiqi Wang, Hanwei Zhu, Yizong Wang, Chuanmin Jia, and Siwei Ma. 2018. MindWorldBench: Evaluating Mental-State-to-Behavior Reasoning in Image-to-Video Generation. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX).* ACM, New York, NY, USA, 8 pages. https://doi.org/XXXXXXX.XXXXXXX

1 Introduction

Recent image-to-video (I2V) models excel at synthesizing realistic and temporally coherent dynamics [13, 16, 21, 32, 39]. However, physical realism does not equate to understanding human behavior. Real-world actions are largely driven by unobservable mental states such as perception, belief, and desire. In cognitive science, the ability to reason about such latent states is referred to as Theory of Mind (ToM) [28]. As shown in Figure 1 (Left), a cognitively capable model must prioritize a person's *subjective belief* over objective reality. For instance, in a false-belief scenario, the model should generate the man opening an empty box because he believes the object is inside, regardless of where the object actually is. Motivated by this, we introduce **mental-state-to-behavior reasoning**, where visual actions emerge from latent psychological states rather than explicit directives.

This cognitive dimension remains absent from existing evaluations. As contrasted in Figure 1 (Middle), current benchmarks predominantly assess explicit action alignment (e.g., "a man opens

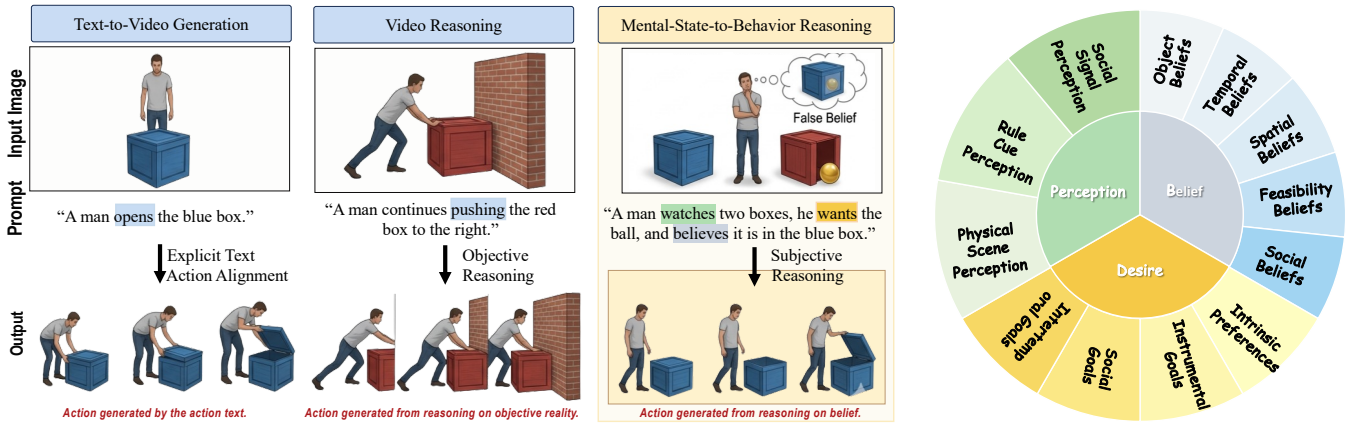


Figure 1: Mental-State-to-Behavior Reasoning and the MindWorldBench Taxonomy. (Left) Evolution of Generation Paradigms: Unlike traditional Text-to-Video (explicit instruction alignment) or Video Reasoning (objective physical rules), our paradigm requires models to autonomously deduce behaviors from latent cognitive states. As illustrated, the model must prioritize the man's *subjective belief* over objective reality to generate the correct reasoning-driven action. (Right) MindWorldBench Taxonomy: Grounded in the BDI-P framework, we systematically categorize mental variables into three primary dimensions—*Perception*, *Belief*, and *Desire*—to comprehensively evaluate Theory of Mind capabilities in video generation.

a box") [14] or implicit physical continuation (e.g., predicting a shot's trajectory) [25]. These settings focus on observable kinematics and fail to isolate mental reasoning from statistical shortcut learning [23]. Models often succeed by memorizing physical priors or associating action verbs with visual outcomes. To gauge the cognitive depth of I2V models, we must shift from physical instruction-following toward mental-state-conditioned behavior generation.

To address this gap, we introduce **MindWorldBench**, the first comprehensive benchmark evaluating ToM capabilities in video generation. We conceptualize this task as a physical-mental decoupling mechanism where the image anchors the *World State* and the text injects the subjective *Mental State*. Grounded in the Belief-Desire-Intention (BDI) theory, our benchmark systematically organizes latent variables into a structured taxonomy of *Perception*, *Belief*, and *Desire* (Figure 1, Right). Crucially, MindWorldBench pioneers a **"Zero-Action Prompting"** strategy. Prompts exclusively describe the character's internal state without action verbs, forcing models to autonomously deduce logical intent and generate corresponding behaviors.

Distinguishing true causal inference from random guessing is a fundamental challenge. We tackle this using a **counterfactual paired design**. By constructing minimal contrastive pairs of mental prompts (e.g., alternating a belief from *true* to *false*) for an identical image, any behavioral divergence provides evidence of causal sensitivity [7, 35]. For scalable assessment, we develop an LVM-based automated pipeline [5, 12, 17, 20, 44] augmented with Chain-of-Thought reasoning [17, 31, 37] to evaluate Video Quality, Commonsense Plausibility, and Mental Reasoning, achieving high correlation with human judgments.

The evaluation is conducted on the MindWorldBench dataset, comprising 744 expert-designed prompts. We evaluate 8 state-of-the-art models including Veo 3.1, Kling V3, Wan 2.6, and CogVideoX. Results reveal a dichotomy where models maintain physical common sense but struggle with mental reasoning. Notably, we identify an **"Omniscient Bias"**. Models frequently fail to decouple their global visual knowledge from the character's limited perception, generating actions based on objective truth rather than subjective belief.

In summary, our main contributions are threefold:

- **A Paradigm Shift in Evaluation:** We propose MindWorldBench to transition from physical instruction-following to Mental-State-to-Behavior reasoning via BDI theory and Zero-Action Prompting.
- **Rigorous Causal Methodology:** We construct a decoupled dataset of 372 images and 744 prompts. The counterfactual design isolates latent mental variables and eliminates statistical shortcuts.
- **Robust Evaluation and Deep Insights:** We establish an LVM-based framework that uncovers cognitive bottlenecks in 8 leading models, identifying the "Omniscient Bias" as a critical challenge for future world simulators.

2 Related Work

Video Generation Models and World Simulators The landscape of video generation has been transformed by diffusion models and Diffusion Transformers [30, 39]. Recent models demonstrate strong capabilities in generating high-fidelity, temporally coherent videos. Excelling at physical dynamics like fluid motion and object permanence, they are increasingly viewed as nascent "world simulators" [2, 21, 43]. However, real-world dynamics are governed by both physics and cognitive intent. While current models successfully simulate the physical consequences of actions, they lack mechanisms to model *why* these actions are initiated.

Evaluation Benchmarks for Video Generation Numerous benchmarks evaluate video generation models, including VBench series

Unpublished working draft. Not for distribution.

[14, 15, 45], EvalCrafter [23], UI2V-Bench [42] and FETV [24]. These mainly focus on perceptual quality and semantic alignment [22, 45]. By testing if generated videos execute explicit actions described in prompts, they adopt an instruction-following paradigm [3]. This encourages statistical shortcut learning because models map textual verbs to visual priors without understanding causal logic. In contrast, MindWorldBench introduces Zero-Action Prompting and counterfactual testing, shifting the focus from simple action execution to deep causal reasoning [6].

Evaluation Dimensions Theory of Mind (ToM) is the cognitive ability to attribute mental states to oneself and others [1]. In AI, ToM is extensively studied using the Belief-Desire-Intention (BDI) framework [8, 9, 27] and textual False Belief tasks for language models [19, 34]. In the visual domain, emerging works explore ToM in vision models but restrict their focus to understanding tasks like VideoQA [4, 10, 20, 40]. The inverse generation task of synthesizing visual behaviors conditioned on latent mental states remains unexplored. MindWorldBench fills this gap by extending the BDI framework into a vision-oriented architecture, establishing the first ToM benchmark for video generation.

3 Method

3.1 Task Definition

We formalize *mental-state-conditioned video generation* at two levels: a latent cognitive reasoning level and a video generation level, as the causal graph illustrated in Figure 2.

Theory of Mind Modeling. As shown in the upper part of Figure 2, human behavior is modeled through a latent causal chain grounded in the BDI-P framework. Given an initial world state W_t, the internal *Belief* (B) is formed from observations and may be constrained by an optional *Perception* variable (P), which is introduced only in scenarios involving partial observability or attentional failure. The *Desire* (D) and *Belief* (B) jointly determine an unobservable *Intention* (I), which then leads to a visible *Action* (A). Together with the physical context W_t, this action determines the subsequent world state W_{t+1}:

$$p(W_{t+1} \mid W_t, \mathcal{M}) = p(W_{t+1} \mid W_t, A)\, p(A \mid I)\, p(I \mid \mathcal{M}) \tag{1}$$

Video Reasoning Paradigm. At the benchmark level, we operationalize this cognitive process as an end-to-end generation task. As shown in the lower part of Figure 2, the latent chain is simplified into a direct mapping from the initial image and mental constraints to the generated video V:

$$(W_t, \mathcal{M}) \xrightarrow{\text{Reasoning}} V, \qquad \mathcal{M} \subseteq \{P, B, D\} \tag{2}$$

where W_t is anchored by the input image, and $\mathcal{M}$ defines the mental-state reasoning space. This mapping requires the model to implicitly infer the latent chain from mental state to intention to action. Accordingly,

$$p(V \mid W_t, \mathcal{M}) = p(V \mid W_t, A)\, p(A \mid I)\, p(I \mid \mathcal{M}) \tag{3}$$

Unlike traditional action-driven paradigms that provide the action A explicitly, MindWorldBench provides only the high-level mental constraints $\mathcal{M}$. The benchmark therefore evaluates whether a model can map invisible mental states to visible behavioral outcomes.

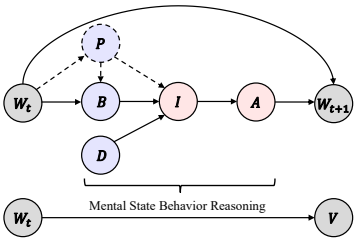


Figure 2: Causal Graph of Mental-State-to-Behavior Generation. Shaded gray nodes (W_t, V) are observed visual states. Blue nodes ($\mathcal{M} = \{P, B, D\}, I, A$) are latent cognitive variables. Models implicitly infer the latent intention I and action A to generate the video V.

3.2 Mental-Space Taxonomy

Grounded in the BDI paradigm and Theory of Mind, our taxonomy parameterizes latent cognitive variables across 3 primary dimensions, 13 secondary aspects, and 34 fine-grained sub-dimensions (Figure 1). We structure this hierarchical space as a reasoning pipeline where perceptual constraints inform mental representations:

Perception (P): The Information and Attentional Gateway. Perception serves as the cognitive gateway between the environment and internal knowledge, involving information acquisition and attentional allocation. Spanning environmental, rule, and social cues, this dimension evaluates whether models recognize that the presence of information does not guarantee character awareness. It tests the ability to synthesize reactive behaviors based on successful cognitive uptake versus inattentional blindness.

Belief (B): The Subjective World Model. Belief represents the character's internal understanding, which may diverge from objective reality. We organize this dimension into physical, temporal, and spatial aspects to test whether models resist omniscient bias. Success requires generating behaviors dictated by the character's potentially flawed or outdated understanding instead of the globally true state.

Desire (D): The Motivational Driver. Desire parameterizes the goals and preferences that drive action selection. Categorized into immediate preference, functional utility, and social motivation, this dimension assesses the capacity to synthesize goal-directed actions. This ensures generated trajectories are causally aligned with internal utilities and specific target states.

3.3 Dataset Construction

MindWorldBench is constructed through a psychology-inspired pipeline grounded in Theory-of-Mind and BDI-style reasoning. We derive mental-state scenarios from classic cognitive settings and instantiate them using publicly available data or AIGC images. Human experts then design task instances with controlled mental-state conditions, candidate actions, and world-state constraints, ensuring each example isolates a specific reasoning factor while remaining visually plausible.

The dataset is organized into 3 primary dimensions and 12 secondary categories, spanning diverse fine-grained mental-state conditions across Perception, Belief, and Desire. This design enables

Unpublished working draft. Not for distribution.

Figure 3: Representative examples from MindWorldBench. The benchmark spans the three mental dimensions: Belief, Desire, and Perception, covering diverse subcategories. Each example includes an input image and a structured mental-state prompt, followed by a question probing expected actions, highlighting the need for models to infer behavior from latent mental states.

controlled evaluation over diverse cognitive factors. Full details are provided in the **Appendix**.

3.4 Evaluation Protocol

To determine whether a generated video reflects genuine mental-state-conditioned reasoning, we adopt an evaluation protocol that jointly assesses video reliability, commonsense validity, behavioral correctness, and world-state consistency.

Evaluation Metrics. We evaluate each generated video using five complementary metrics. **Visual Quality** measures perceptual quality on a 1–5 scale, including consistency with the input image, preservation of character and scene identity, clarity, temporal stability, and motion naturalness. This metric ensures that reasoning judgments are not confounded by severe generation failures. **Commonsense Plausibility** evaluates whether the generated behavior and its consequences are compatible with basic physical and commonsense knowledge, also on a 1–5 scale. **Action Occurrence** identifies which action appears in the video: one of the two intention-related candidate actions, or an irrelevant *other action*. This metric tests whether the model responds to the mental-state manipulation with a task-relevant behavior. **Intention Accuracy** evaluates whether the generated action is consistent with the intention implied by the mental-state condition. For example, if a character falsely believes a ball is in the left box and wants to retrieve it, the correct behavior is to open the *left* box, following the character's belief rather than the true world state. **World-State Maintenance** evaluates whether the visible environment still follows the objective world state after the action is performed. In the same example, once the left box is opened, the video should still show that it is empty, since the ball remains in the right box. This metric tests whether the model preserves the distinction between subjective mental state and objective physical reality.

LVM-based Evaluation and Validation. We use Gemini 3.1 Pro as the evaluator and adopt a structured chain-of-thought judging pipeline. For each test case, the evaluator is guided to identify the objective world state, the manipulated mental-state condition, and the distinction between the candidate actions, and then score the generated video using detailed rubrics for the five metrics above. To reduce scoring instability, we provide metric-specific guidelines and in-context examples covering both positive and negative cases. We further compare LVM-based judgments with human expert annotations on a subset of benchmark samples and observe high agreement, suggesting that the proposed pipeline provides stable, human-aligned assessments at benchmark scale.We validate the reliability of the LVM-based evaluation through a user study by comparing model-generated scores with human annotations on a subset of benchmark samples. The results show strong correlation between the two, indicating that the automated evaluation is well aligned with human judgment. Details of the study design and analysis are provided in the **Appendix**.

4 Experiments

4.1 Experimental Setup

To assess the cognitive reasoning capabilities of current video generation models, we show eight state-of-the-art Image-to-Video (I2V) systems, spanning a mix of closed-source commercial APIs and prominent open-source foundation models: Veo 3.1 [38], Kling V3 [33], Seedance 1.5 Pro [30], Hailuo 2.3 [26], Runway Gen4 [29], Wan 2.6 [36], Wan 2.2 [36], and CogVideoX [41]. The evaluation is conducted strictly on the MindWorldBench dataset, which comprises 744 expert-designed mental-state prompts. To support a fair

Unpublished working draft. Not for distribution.

Table 1: Overall performance comparison on MindWorldBench.

Models	Params	Visual Quality ↑	Commonsense Plausibility ↑	Intention Accuracy ↑	World-State Maintenance ↑
Kling [33]	-	4.1	4.2	47.2	66.3
Veo3.1 [38]	-	3.8	4.0	59.5	61.4
Seedance1.5Pro [30]	-	4.3	4.3	29.3	52.9
Gen4 [29]	-	4.0	3.9	18.6	49.4
Hailuo2.3 [26]	-	3.8	3.8	13.0	32.5
CogVideoX [41]	5 B	3.9	4.0	27.7	54.2
Wan2.6 [36]	14 B	4.2	4.3	39.0	59.9
Wan2.2 [36]	14 B	4.4	4.4	28.7	54.0

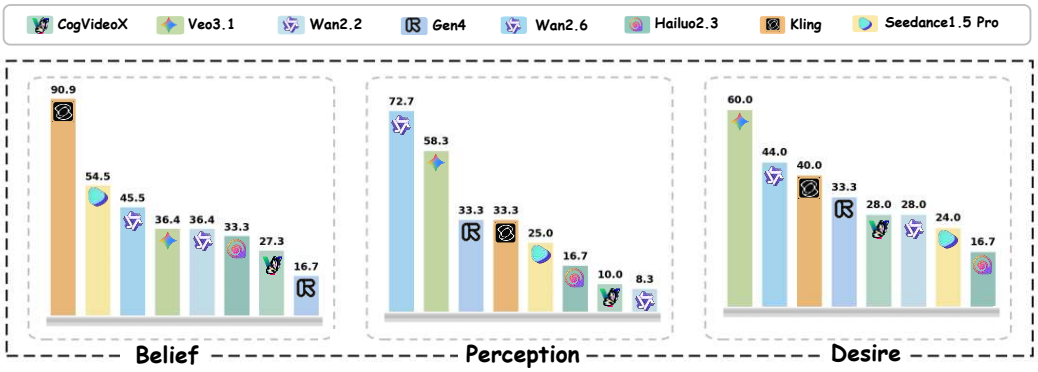


Figure 4: Performance breakdown across the three major mental dimensions: Belief, Perception, and Desire. Different models exhibit distinct strengths across dimensions.

comparison across diverse architectures, we align generation configurations to the closest natively supported equivalents for each model. All models are evaluated under the zero-action I2V setting, targeting 720p resolution and a video duration of 5 to 8 seconds. More open-sourced results of HunyuanI2V [18] and LTX2.3 [11] are in **Appendix**.

4.2 Overall Benchmark Results

Table 1 reports the overall performance of all models on MindWorldBench. Across models, the two perceptual metrics are relatively strong and tightly clustered: Visual Quality ranges from 3.8 to 4.4, and Commonsense Plausibility ranges from 3.8 to 4.4. In contrast, the two reasoning-related metrics show much larger variation, with Intention Accuracy ranging from 13.0% to 59.5% and World-State Maintenance ranging from 32.5% to 66.3%. This gap suggests that current image-to-video models are better at generating visually plausible videos than at inferring behavior from latent mental states and preserving the objective world state after action execution.

No model dominates all four metrics. Wan2.2 and Seedance-1.5 Pro achieve the strongest perceptual quality, while Veo3.1 obtains the highest Intention Accuracy (59.5%), and Kling achieves the best World-State Maintenance (66.3%). Wan2.6 remains competitive on both perceptual metrics and world-state preservation, whereas Hailuo2.3 performs weakest on both reasoning-related metrics. Overall, these results reveal a trade-off between perceptual fidelity and mental-state reasoning: better-looking videos do not necessarily imply better intention inference or world-state preservation.

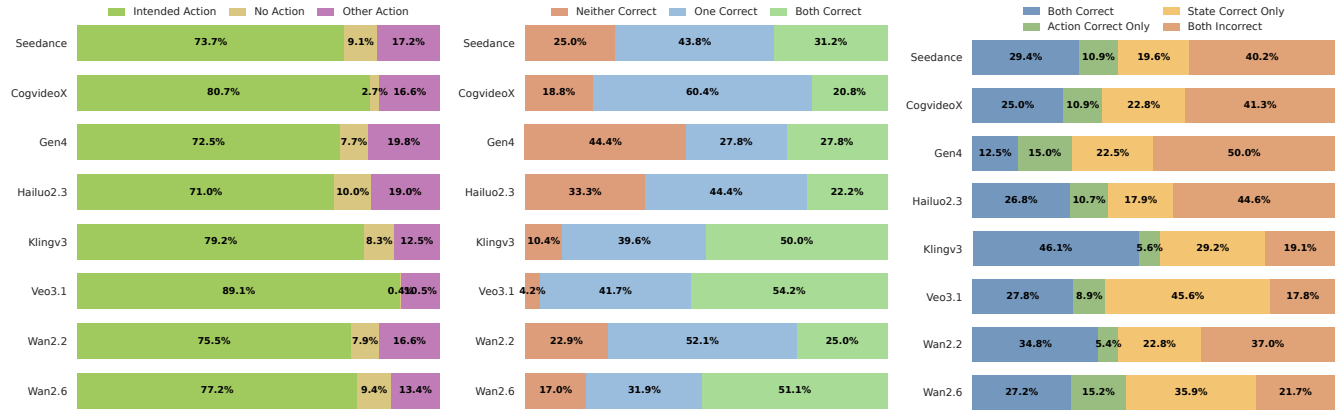
Figure 4 further breaks down model performance across the three major mental dimensions: Belief, Perception, and Desire. The results show clear dimension-specific strengths. Kling performs best on Belief, Wan2.6 leads on Perception, and Veo3.1 achieves the highest score on Desire. This pattern indicates that current models do not exhibit uniform reasoning capability, but instead show uneven strengths across different types of mental-state inference. No model consistently ranks first across all three dimensions, highlighting the importance of fine-grained evaluation beyond overall averages.

4.3 Counterfactual Consistency Analysis

To investigate whether models genuinely parse latent mental states rather than potentially relying on spurious visual correlations, we analyze their performance on counterfactual pairs. In each pair, the initial physical scene is identical, while the underlying mental condition is minimally altered. We define a *Both Correct* outcome as the model successfully generating the intended action in both contrasting scenarios, serving as a stricter proxy for causal sensitivity to mental-state interventions.

Figure 5b visualizes the distribution of these paired outcomes. Across all systems, the rate of joint success (*Both Correct*) remains consistently low. Instead, the distribution is heavily skewed toward the *Only One Correct* outcome (e.g., 60.4% for CogVideoX and 52.1% for Wan 2.2). **This asymmetric distribution strongly suggests a reliance on shortcut learning rather than robust causal reasoning**: models frequently succeed when the prompt aligns

Unpublished working draft. Not for distribution.



(a) Action Types **(b) Counterfactual Outcomes** **(c) Action–State Decomposition**

Figure 5: Diagnostic Analysis of Reasoning Failures. (a) Generated action type distribution. (b) Counterfactual paired outcomes. (c) Decomposition of action and world-state results into Both Correct, Action Correct, State Correct, and Both Incorrect.

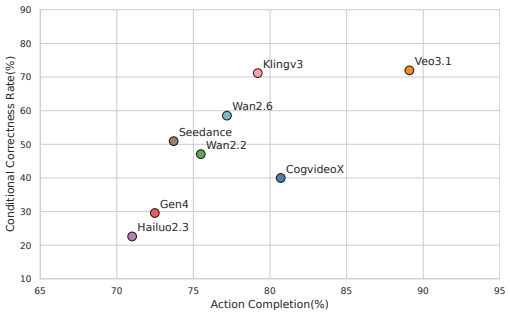


Figure 6: Action Completion (x-axis) vs. Conditional Correctness (y-axis), illustrating the trade-off between action generation and reasoning accuracy.

with expected physical priors, but appear to struggle when a minimal counterfactual modification requires them to override those learned associations.

Furthermore, the substantial proportion of *Both Wrong* outcomes across several models highlights a broader fragility under paired intervention. Collectively, these findings support the utility of our counterfactual evaluation protocol: single-prompt accuracy may frequently overestimate cognitive capabilities, whereas counterfactual consistency provides a more reliable lens to separate mental-state tracking from superficial pattern matching.

4.4 Action Realization and Action–State Consistency

Action Type Distribution. A model must generate task-relevant behaviors to succeed. Figure 5a shows the proportion of outputs categorized as *Intended Action*, *No Action*, or *Other Action*. The data indicates most models frequently produce intended actions (71.0%–89.1%), while rendering failures like *No Action* remain rare (0.4%–10.0%). This suggests the observed low reasoning accuracy is likely driven by cognitive bottlenecks in action selection, rather than an inability to render dynamic motions.

Conditional Correctness and Action–State Decomposition. Figure 6 contextualizes this capability by plotting the completion rate of intended actions against their conditional correctness. Veo 3.1 and Kling V3 demonstrate a balanced capacity, achieving both high completion and higher correctness. In contrast, CogVideoX and Wan 2.2 exhibit high completion but moderate correctness, appearing to act as capable motion generators that still struggle with accurate intent inference. Gen4 and Hailuo 2.3 underperform on both axes.

Furthermore, Figure 5c decomposes outputs into *Both Correct*, *Action Correct Only*, *State Correct Only*, and *Both Incorrect*. Achieving joint success remains challenging; Kling V3 reaches 46.1%, while Gen4 (12.5%) and Wan 2.6 (27.2%) struggle to satisfy both constraints simultaneously. Notably, an asymmetric error profile emerges: *State Correct Only* consistently surpasses *Action Correct Only*. This implies that maintaining an observable objective world state may be inherently less demanding than synthesizing actions conditioned on latent mental variables. These findings suggest action reasoning and world-state maintenance are distinct capabilities, and failures stem from both.

5 Conclusion

We present MindWorldBench, a benchmark for evaluating mental-state-conditioned video generation. By introducing a structured mental-space taxonomy and a controlled counterfactual design, it enables systematic assessment of models' sensitivity to latent mental variables such as belief, desire, and perception. Experiments show that although current video generation models achieve strong visual quality and basic commonsense plausibility, they still struggle with mental reasoning, often failing to produce intention-consistent actions or preserve the correct world state. These results reveal a clear gap between visual realism and cognitive reasoning, suggesting the need for explicit modeling of agent cognition in future world simulators.

Unpublished working draft. Not for distribution.

References

[1] Michael Bratman. 1987. Intention, plans, and practical reason. (1987).

[2] Tim Brooks, Bill Peebles, Conor Holmes, Will DePue, Craig Donahue, Aditya Ramesh, et al. 2024. Video Generation Models as World Simulators. https://openai.com/index/video-generation-models-as-world-simulators/. OpenAI technical report, accessed 2026-04-02.

[3] Zefan Cai, Haoyi Qiu, Tianyi Ma, Haozhe Zhao, Gengze Zhou, Kung-Hsiang Huang, Parisa Kordjamshidi, Minjia Zhang, Wen Xiao, Jiuxiang Gu, et al. 2025. MMGR: Multi-Modal Generative Reasoning. *arXiv preprint arXiv:2512.14691* (2025).

[4] Dongping Chen, Ruoxi Chen, Shilin Zhang, Yaochen Wang, Yinuo Liu, Huichi Zhou, Qihui Zhang, Yao Wan, Pan Zhou, and Lichao Sun. 2024. Mllm-as-a-judge: Assessing multimodal llm-as-a-judge with vision-language benchmark. In *Forty-first International Conference on Machine Learning*.

[5] Xiuyuan Chen, Yuan Lin, Yuchen Zhang, and Weiran Huang. 2024. Autoeval-video: An automatic benchmark for assessing large vision language models in open-ended video question answering. In *European Conference on Computer Vision*. Springer, 179–195.

[6] Yuefei Chen, Jiang Liu, Xiaodong Lin, and Ruixiang Tang. 2025. CounterVQA: Evaluating and Improving Counterfactual Reasoning in Vision-Language Models for Video Understanding. *arXiv preprint arXiv:2511.19923* (2025).

[7] Jon Gauthier, Jennifer Hu, Ethan Wilcox, Peng Qian, and Roger Levy. 2020. SyntaxGym: An online platform for targeted evaluation of language models. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: System Demonstrations*. 70–76.

[8] Michael Georgeff, Barney Pell, Martha Pollack, Milind Tambe, and Michael Wooldridge. 1998. The belief-desire-intention model of agency. In *International workshop on agent theories, architectures, and languages*. Springer, 1–10.

[9] M Georgeff and A Rao. 1991. Modeling rational agents within a BDI-architecture. In *Proc. 2nd Int. Conf. on Knowledge Representation and Reasoning (KR'91). Morgan Kaufmann*. of, 473–484.

[10] Jiawei Gu, Xuhui Jiang, Zhichao Shi, Hexiang Tan, Xuehao Zhai, Chengjin Xu, Wei Li, Yinghan Shen, Shengjie Ma, Honghao Liu, et al. 2024. A survey on llm-as-a-judge. *The Innovation* (2024).

[11] Yoav HaCohen, Benny Brazowski, Nisan Chiprut, Yaki Bitterman, Andrew Kvochko, Avishai Berkowitz, Daniel Shalem, Daphna Lifschitz, Dudu Moshe, Eitan Porat, Eitan Richardson, Guy Shiran, Itay Chachy, Jonathan Chetboun, Michael Finkelson, Michael Kupchick, Nir Zabari, Nitzan Guetta, Noa Kotler, Ofir Bibi, Ori Gordon, Poriya Panet, Roi Benita, Shahar Armon, Victor Kulikov, Yaron Inger, Yonatan Shiftan, Zeev Melumian, and Zeev Farbman. 2025. LTX-2: Efficient Joint Audio-Visual Foundation Model. *arXiv preprint arXiv:2601.03233* (2025).

[12] Hui Han, Siyuan Li, Jiaqi Chen, Yiwen Yuan, Yuling Wu, Yufan Deng, Chak Tou Leong, Hanwen Du, Junchen Fu, Youhua Li, et al. 2025. Video-bench: Human-aligned video generation benchmark. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. 18858–18868.

[13] Haoyang Huang, Guoqing Ma, Nan Duan, Xing Chen, Changyi Wan, Ranchen Ming, Tianyu Wang, Bo Wang, Zhiying Lu, Aojie Li, et al. 2025. Step-video-ti2v technical report: A state-of-the-art text-driven image-to-video generation model. *arXiv preprint arXiv:2503.11251* (2025).

[14] Ziqi Huang, Yinan He, Jiashuo Yu, Fan Zhang, Chenyang Si, Yuming Jiang, Yuanhan Zhang, Tianxing Wu, Qingyang Jin, Nattapol Chanpaisit, et al. 2024. Vbench: Comprehensive benchmark suite for video generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 21807–21818.

[15] Ziqi Huang, Fan Zhang, Xiaojie Xu, Yinan He, Jiashuo Yu, Ziyue Dong, Qianli Ma, Nattapol Chanpaisit, Chenyang Si, Yuming Jiang, et al. 2025. Vbench++: Comprehensive and versatile benchmark suite for video generative models. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).

[16] Sai Shashank Kalakonda, Shubh Maheshwari, and Ravi Kiran Sarvadevabhatla. 2025. Morag-Multi-Fusion Retrieval Augmented Generation for Human Motion. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 4564–4573.

[17] Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. 2022. Large language models are zero-shot reasoners. *Advances in neural information processing systems* 35 (2022), 22199–22213.

[18] Weijie Kong, Qi Tian, Zijian Zhang, Rox Min, Zuozhuo Dai, Jin Zhou, Jiangfeng Xiong, Xin Li, Bo Wu, Jianwei Zhang, et al. 2024. Hunyuanvideo: A systematic framework for large video generative models. *arXiv preprint arXiv:2412.03603* (2024).

[19] Michal Kosinski. 2024. Evaluating large language models in theory of mind tasks. *Proceedings of the National Academy of Sciences* 121, 45 (2024), e2405460121.

[20] Jian Li, Weiheng Lu, Hao Fei, Meng Luo, Ming Dai, Min Xia, Yizhang Jin, Zhenye Gan, Ding Qi, Chaoyou Fu, et al. 2024. A survey on benchmarks of multimodal large language models. *arXiv preprint arXiv:2408.08632* (2024).

[21] Shaowei Liu, Zhongzheng Ren, Saurabh Gupta, and Shenlong Wang. 2024. Physgen: Rigid-body physics-grounded image-to-video generation. In *European Conference on Computer Vision*. Springer, 360–378.

[22] Xinxin Liu, Zhaopan Xu, Ming Li, Kai Wang, Yong Jae Lee, and Yuzhang Shang. 2025. Can World Simulators Reason? Gen-ViRe: A Generative Visual Reasoning Benchmark. *arXiv preprint arXiv:2511.13853* (2025).

[23] Yaofang Liu, Xiaodong Cun, Xuebo Liu, Xintao Wang, Yong Zhang, Haoxin Chen, Yang Liu, Tieyong Zeng, Raymond Chan, and Ying Shan. 2024. Evalcrafter: Benchmarking and evaluating large video generation models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 22139–22149.

[24] Yuanxin Liu, Lei Li, Shuhuai Ren, Rundong Gao, Shicheng Li, Sishuo Chen, Xu Sun, and Lu Hou. 2023. Fetv: A benchmark for fine-grained evaluation of open-domain text-to-video generation. *Advances in Neural Information Processing Systems* 36 (2023), 62352–62387.

[25] Fanqing Meng, Jiaqi Liao, Xinyu Tan, Wenqi Shao, Quanfeng Lu, Kaipeng Zhang, Yu Cheng, Dianqi Li, Yu Qiao, and Ping Luo. 2024. Towards world simulator: Crafting physical commonsense-based benchmark for video generation. *arXiv preprint arXiv:2410.05363* (2024).

[26] MiniMax. 2025. Hailuo 2.3. https://hailuoai.video/zh-Intl. Accessed: 2026-04-02.

[27] Michael C Móra, José G Lopes, Rosa M Viccariz, and Helder Coelho. 1998. BDI models and systems: Reducing the gap. In *International Workshop on Agent Theories, Architectures, and Languages*. Springer, 11–27.

[28] David Premack and Guy Woodruff. 1978. Does the chimpanzee have a theory of mind? *Behavioral and brain sciences* 1, 4 (1978), 515–526.

[29] Runway. 2025. Introducing Runway Gen-4. https://runwayml.com/research/introducing-runway-gen-4 Accessed: 2026-04-02.

[30] Team Seedance, Heyi Chen, Siyan Chen, Xin Chen, Yanfei Chen, Ying Chen, Zhuo Chen, Feng Cheng, Tianheng Cheng, Xinqi Cheng, et al. 2025. Seedance 1.5 pro: A Native Audio-Visual Joint Generation Foundation Model. *arXiv preprint arXiv:2512.13507* (2025).

[31] Hao Shao, Shengju Qian, Han Xiao, Guanglu Song, Zhuofan Zong, Letian Wang, Yu Liu, and Hongsheng Li. 2024. Visual cot: Advancing multi-modal language models with a comprehensive dataset and benchmark for chain-of-thought reasoning. *Advances in Neural Information Processing Systems* 37 (2024), 8612–8642.

[32] Xiaoyu Shi, Zhaoyang Huang, Fu-Yun Wang, Weikang Bian, Dasong Li, Yi Zhang, Manyuan Zhang, Ka Chun Cheung, Simon See, Hongwei Qin, et al. 2024. Motion-i2v: Consistent and controllable image-to-video generation with explicit motion modeling. In *ACM SIGGRAPH 2024 Conference Papers*. 1–11.

[33] Kling Team, Jialu Chen, Yuanzheng Ci, Xiangyu Du, Zipeng Feng, Kun Gai, Sainan Guo, Feng Han, Jingbin He, Kang He, et al. 2025. Kling-Omni Technical Report. *arXiv preprint arXiv:2512.16776* (2025).

[34] Tomer Ullman. 2023. Large language models fail on trivial alterations to theory-of-mind tasks. *arXiv preprint arXiv:2302.08399* (2023).

[35] Jannis Vamvas and Rico Sennrich. 2021. On the limits of minimal pairs in contrastive evaluation. In *Proceedings of the Fourth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*. 58–68.

[36] Team Wan, Ang Wang, Baole Ai, Bin Wen, Chaojie Mao, Chen-Wei Xie, Di Chen, Feiwu Yu, Haiming Zhao, Jianxiao Yang, et al. 2025. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314* (2025).

[37] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. 2022. Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* 35 (2022), 24824–24837.

[38] Thaddäus Wiedemer, Yuxuan Li, Paul Vicol, Shixiang Shane Gu, Nick Matarese, Kevin Swersky, Been Kim, Priyank Jaini, and Robert Geirhos. 2025. Video models are zero-shot learners and reasoners. *arXiv preprint arXiv:2509.20328* (2025).

[39] Zhen Xing, Qijun Feng, Haoran Chen, Qi Dai, Han Hu, Hang Xu, Zuxuan Wu, and Yu-Gang Jiang. 2024. A survey on video diffusion models. *Comput. Surveys* 57, 2 (2024), 1–42.

[40] Hui Yang, Lekha Chaisorn, Yunlong Zhao, Shi-Yong Neo, and Tat-Seng Chua. 2003. VideoQA: question answering on news video. In *Proceedings of the eleventh ACM international conference on Multimedia*. 632–641.

[41] Zhuoyi Yang, Jiayan Teng, Wendi Zheng, Ming Ding, Shiyu Huang, Jiazheng Xu, Yuanming Yang, Wenyi Hong, Xiaohan Zhang, Guanyu Feng, et al. 2024. Cogvideox: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072* (2024).

[42] Ailing Zhang, Lina Lei, Dehong Kong, Zhixin Wang, Jiaqi Xu, Fenglong Song, Chun-Le Guo, Chang Liu, Fan Li, and Jie Chen. 2025. UI2V-Bench: An Understanding-based Image-to-video Generation Benchmark. *arXiv preprint arXiv:2509.24427* (2025).

[43] Chenyu Zhang, Daniil Cherniavskii, Antonios Tragoudaras, Antonios Vozikis, Thijmen Nijdam, Derck WE Prinzhorn, Mark Bodracska, Nicu Sebe, Andrii Zadaianchuk, and Efstratios Gavves. 2025. Morpheus: Benchmarking physical reasoning of video generative models with real physical experiments. *arXiv preprint arXiv:2504.02918* (2025).

[44] Zhuosheng Zhang, Aston Zhang, Mu Li, Hai Zhao, George Karypis, and Alex Smola. 2023. Multimodal chain-of-thought reasoning in language models. *arXiv preprint arXiv:2302.00923* (2023).

Unpublished working draft. Not for distribution.

[45] Dian Zheng, Ziqi Huang, Hongbo Liu, Kai Zou, Yinan He, Fan Zhang, Lulu Gu, Yuanhan Zhang, Jingwen He, Wei-Shi Zheng, et al. 2025. Vbench-2.0: Advancing video generation benchmark suite for intrinsic faithfulness. *arXiv preprint arXiv:2503.21755* (2025).

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009

Unpublished working draft.
Not for distribution.